One loop, five versions

IEEE SENSORS 2026 tutorial, Sunday 25 October: From Naïve Loops to FAIR Datasheets

One simulated soft sensor, one indentation staircase, one acquisition loop. Each version fixes the bug the previous one left behind.

What the five versions taught

	Loop	What it taught
V0	Blocking move, then sleep	Nothing is logged during ramps and holds.
V1	Non-blocking polling	Coverage is fixed. Disk stalls give jitter (p99 about 60 ms).
V2	Threads and a first-in, first-out queue	Jitter is fixed. Rows are stamped when read, so they lag the true time by 14 to 21 ms. At an unlucky start phase it loses about a third of the frames, and its log cannot show it.
V3	Stamp at frame time, add a sequence number	Lag drops to about 0. The loss is now visible in the log.
V4	Coordinator tick at 2× the frame rate	Loss is at most 1.2 % at any start phase, and every lost frame shows in the log.

Whatever the testing path does not record, no analysis can recover. A test that passes is not the claim made true.

Run the demo yourself

On your laptop. Get the repo ([QR]). In the folder that holds `naive_vs_threaded.py`, run:

```
uv run --with numpy --with matplotlib python naive_vs_threaded.py --fast --seed 2
```

`--fast` runs a shorter staircase than the full-length run on the slides. Add `--phase-sweep` to see the loss at start offsets from 0 to 18 ms.

In Colab. Open the notebook ([QR]) and run its cells in order. You need no install.

Why your numbers will differ from the slides

- The sensor and the disk are simulated. The disk-stall model is an assumption, so the jitter figures show the shape of the problem, not your disk.
- Laptop, operating system and system timer change when threads wake up, so the same code gives different jitter and different loss.
- The start phase matters most. The loss in V2 ranges from 0 to 38 % as the start offset moves from 0 to 18 ms. Press Start a few milliseconds later and you get a different datasheet.
- The slides show the full-length run. `--fast` is shorter, so its numbers carry more edge effects.

Compare V2 and V4 on your own machine, then write your two loss figures below.

Your run

Operating system: _____ CPU: _____

V2 loss (%): _____ V4 loss (%): _____